CT-guided Bone Biopsy Planning
Assignment for CISC/CMPE 330

Problem Statement

Percutaneous, or through-the-skin, needle-based procedures are widely used throughout the body. Several such procedures are commonly performed in the pelvic bones, including biopsy, tumor ablation (i.e., destruction of tumor tissue by delivering thermal energy), cementoplasty (i.e., removal of tumor tissue and filling the resulting void with bone cement to relieve pain and reduce the risk of subsequent fracture), and fixation (i.e., prophylactic or post-traumatic reinforcement of the bone using screws).

These procedures may be performed individually or in combination and often rely on the same basic technique: a thin tubular conduit is inserted through the soft tissues to the bone, and the intervention is then performed through this access path. Depending on the procedure, this may involve obtaining a biopsy specimen, delivering thermal energy, injecting bone cement, drilling bone, or inserting fixation screws.



Fig. 1: CT image of the biopsy site. (For clarity, we show here a post-insertion image with the bone coring biopsy needle, and nearby nerves and blood vessels marked in the image.)

In this exercise, we consider an actual clinical case involving a 50-year-old woman who was previously treated for lung cancer and now presents with several new lytic bone lesions and right hip pain. We intend to perform a core-needle biopsy of the bone lesion under intraoperative computed tomography (CT) image guidance (Fig. 1).

The success of the procedure depends to a large extent on accurately guiding the biopsy needle into the target while minimizing collateral injury to surrounding structures. The biopsy needle, typically about 3.0 mm in diameter, must be navigated through a narrow access window in the iliac bone and advanced into the lesion within brittle bone. A tissue sample is then obtained using a coring drill.

In this exercise, we will computationally navigate the biopsy tool within the CT scanner. The tool will be instrumented with three spherical markers that are clearly visible in the CT images and can therefore be used to determine the position and orientation of the tool relative to the CT coordinate frame.

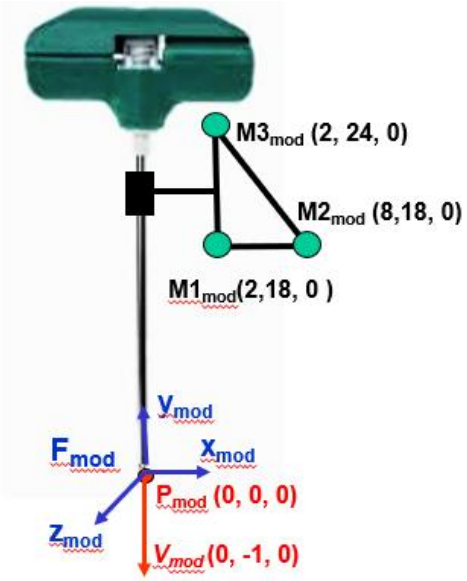


Fig. 2: Model of the "ground truth" tracked tool

Tool Calibration (your task)

Prior to using the tracked tool, it must be calibrated; that is, we need to determine the position of the tool tip and the tool axis with respect to the Cartesian coordinate frame defined by the three markers. Fig. 2 shows an idealized model, or "ground-truth" representation, of the tool. The model includes the three markers, the tool tip, and the tool axis

vector, all expressed in the tool moel frame (F_{mod}). We assume that the three markers are also visible to an optical tracking system, which will provide the observations used for calibration.

Your task is to design and implement software to calibrate both the tool tip and the tool axis using measurements obtained from the optical tracker. Explain your approach step by step, with particular attention to the motion constraints used during calibration. Using in-code comments, and additional document or drawing if helpful, explain your approach step by step.

Test your calibration software on simulated data. Without loss of generality, you may choose the tracker coordinate frame to coincide with the tool model frame. Generate appropriate motions of the tool, compute the corresponding marker positions, provide these simulated observations as input to your calibration software, and demonstrate that the calibration correctly recovers the parameters of the "ground-truth" tool: the tool-tip position and the tool-axis direction vector.

Second, run a simulated test of a more realistic optical tracker by adding normally distributed Gaussian noise to the marker observations. Select the noise level such that, in 95% of observations, the marker-position error is less than 1.0 mm. Repeat the calibration and quantify the resulting errors in the estimated tool-tip position and tool-axis direction. Explain your findings.

Consult the course notes as needed. Depending on the calibration method you choose, useful computational utilities may include coordinate-frame transformations, sphere fitting, circle fitting, plane fitting. Implement such utilities as separate modules where appropriate.

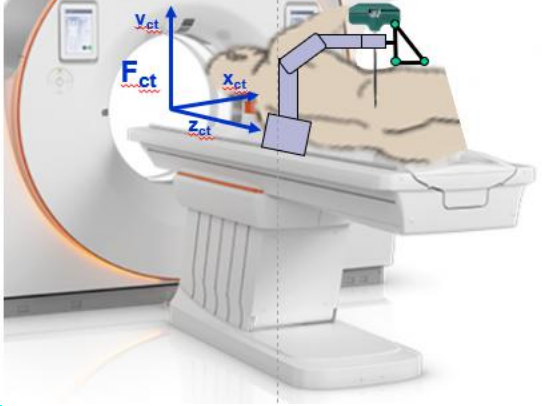


Fig. 3: Patient on the CT table, in treatment position, with the passive needle holder mounted in place.

Biopsy Setup

The intraoperative scene is shown in Fig. 3. The patient is positioned on the CT table in front of the scanner in the left lateral decubitus position, allowing access to the right iliac bone from above. The legs are slightly flexed to improve stability and reduce the risk of rolling on the table.

The patient is then immobilized in a vacuum cushion and secured with Velcro straps to minimize motion. We assume that an adjustable device holds the biopsy tool in a fixed position above the patient.

Imaging & Visualization

After the setup is complete, the patient, together with the tool-holding device are translated into the CT scanner. We acquire a series of parallel cross-sectional CT images, which are then loaded into the treatment-planning software, such as illustrated in Fig. 4. Within this image stack, the physician segments the target structure—in our case, the bone tumor—on each relevant slice by placing a series of points along the tumor boundary. The

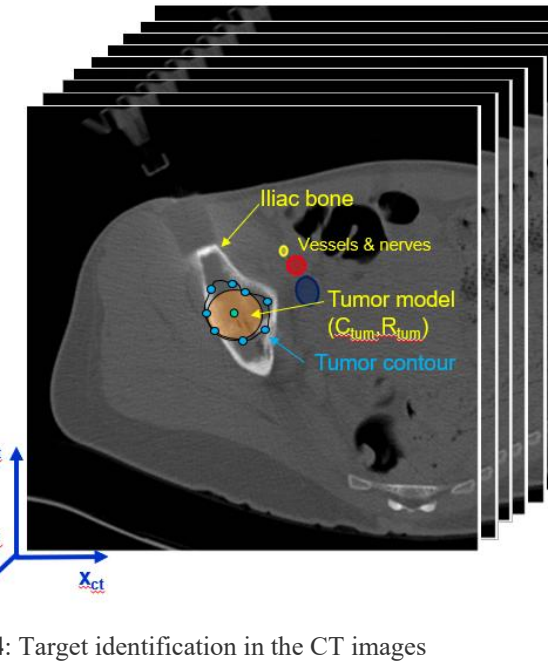


Fig. 4: Target identification in the CT images

segmentation may be performed by simply clicking with a mouse, or with the assistance of advanced machine-learning or AI-based software tools.

From these contour points, we reconstruct a geometric model of the tumor for biopsy planning. In many cases, a simple sphere provides a reasonable approximation of the tumor geometry. To help ensure a successful biopsy, the target model should not overestimate the true extent of the tumor. We therefore often approximate the target using the largest sphere that fits snugly within the segmented tumor volume.

The physician then selects a suitable skin entry point, sterilizes the entry area, places the tip of the biopsy tool at the selected entry point, and aims the tool toward the target, and a new set of CT images is acquired – shown in Fig. 5.

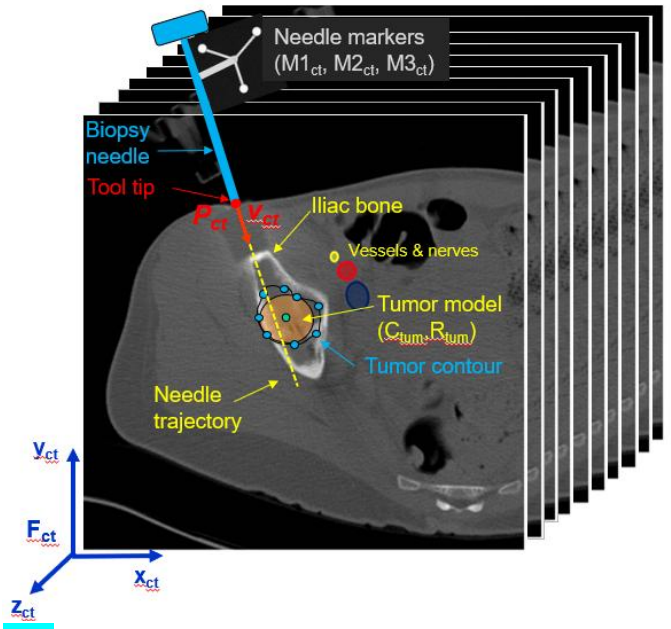


Fig. 5: Digital model of the scene on a computer screen. (Note that this is a mock rendering for this exercise.)

Registration (your task)

For registering the tool to CT coordinate frame, we use three markers rigidly attached to the tool. These markers are identified in the 3D CT image stack and used to register the surgical tool to the CT coordinate frame. For our application, we compute the position of the needle tip and the direction vector of the needle axis in CT coordinates. These quantities are sufficient for planning and guiding the biopsy.

Your task is to design and implement a module to register the tracked needle's tip and tool axis vector to CT imaging frame.

Input information includes the $M1_{tool}$, $M2_{tool}$, $M3_{tool}$ – three markers in tool model frame; P_{tool} – tool tip in tool model frame; v_{tool} – tool axis vector in tool model frame; $M1_{ct}$, $M2_{ct}$, $M3_{ct}$ – three markers in CT frame

Output information includes P_{ct} – tool tip in CT frame; V_{ct} – tool axis vector in CT frame.

For testing, place the ground truth tool model into a known pose in the CT frame, determine outcome on paper, compute the registration, and prove that you reproduced the predicted results. Generate three test cases, one with pure translation, one with pure rotation, and one with both.

Biopsy Plan (your task)

Given the current tool position in the CT coordinate frame, we can compute the salient parameters of the biopsy scene for the clinician's consideration. Once the biopsy plan is satisfactory, then we may proceed with inserting the needle and harvesting a tissue sample. Otherwise, we need to adjust the biopsy tool, reimage the patient, and evaluate the new plan. This process is called "iterative forward planning". (In your next project, you will compute "inverse kinematic plan" for a remotely actuated robot to aim the tool at the target.)

Now, your task is to design and implement software to compute the salient parameters of a biopsy plan. Assume that we modelled tumor with a sphere. The global plan is to insert the needle along its current trajectory, all the way into the tumor. Compute the following for the physician's consideration:

- Targeting error, as the distance of the needle axis from the tumor center.

- Length of the tissue core that we could harvest from the tumor.
- Required needle insertion depth for harvesting this tissue core.

Input information includes:

- [C_{tum}, R_{tum}] – center and radius of the tumor model in CT frame
- [P_{ct}, v_{ct}] – Needle tip position and needle direction vector in CT frame

Output information includes:

- Targeting Error
- Length of Tissue Core
- Needle Insertion Depth

Testing: Construct ground truth test cases for the possible scenarios: needle intersects the tumor, needle touches the tumor, needle entirely misses the tumor. Run the program and examine if your program produces the expected output.

Some hints: Depending on your approach, useful utilities are likely to include Distance-Of-Point-From-Line, Intersect-Line-and-Sphere; implement these utilities as separate modules where appropriate. In creating ground truth test scenarios, as always, you should think of the simplest spatial arrangements that are mathematically sufficiently general. With in mind, you might consider the tumor model as a unit sphere placed anywhere you like, the skin entry point placed anywhere you like. Then, you need to think of some simple needle trajectories for these three scenarios.

Targeting Error Simulation (your task)

The accuracy of the biopsy depends strongly on the accuracy with which the tool markers are localized in the CT image stack. Marker localization is inevitably subject to error, which propagates into the estimated tool-tip position and tool-axis direction and may ultimately cause the needle trajectory to miss the intended target. A second source of error is uncertainty in identifying and segmenting the target itself. Patient motion is yet another major contributor. Until the tool is firmly engaged with the bone, mechanical forces applied during needle insertion may dislocate the target anatomy. These sources of uncertainty must be taken into account before entering the tool into the patient.

The scene that we consider is depicted in Fig. 6. Without loss of mathematical generality, we assume that the F_{mod} tool model frame and the F_{ct} imaging frame are identical. In an ideal world without any error, we would localize the markers at $M1_{CT}$, $M2_{CT}$, $M3_{CT}$ (now the same as $M1_{mod}$, $M2_{mod}$, $M3_{mod}$) and that the tool is accurately aimed at the target point.

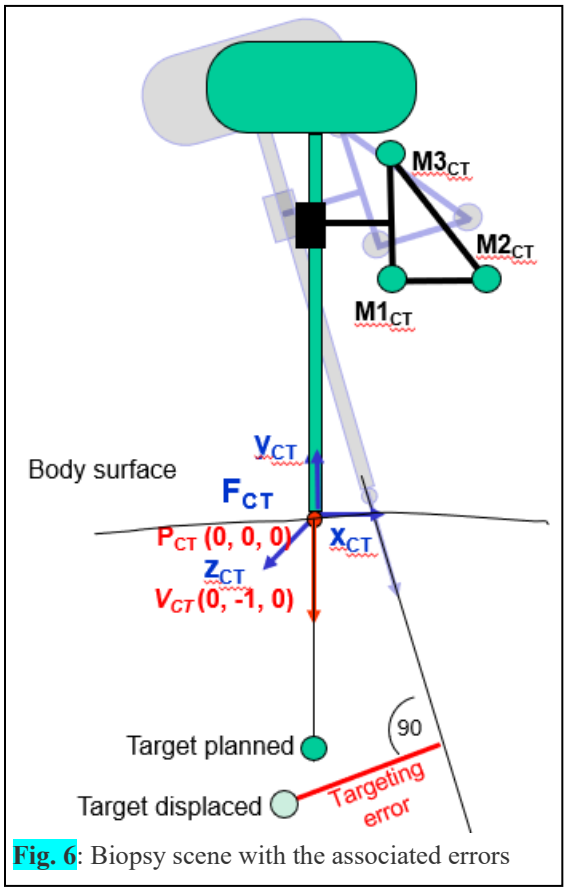


Fig. 6: Biopsy scene with the associated errors

In practice, marker localization is subject to error. As a result, the estimated pose of the tool is slightly inaccurate, leading to errors in both the estimated needle-tip position and the needle-axis direction. At the same time, the target anatomy may be displaced from its planned position due to mechanical forces applied to the patient during manipulation of the tool.

If the needle were inserted under these conditions, its actual trajectory could deviate from the intended target point. We define the resulting distance between the needle trajectory and the displaced target point as the compound targeting error, as it reflects the combined effects of tool-localization error and target displacement.

The actual insertion depth and the length of the tissue core would also deviate from the planned values; however, these effects will be disregarded in this exercise.

We can make the following simplifying assumptions:

- All error sources are random, mutually independent, and modeled as zero-mean Gaussian errors in three dimensions.
- The tracking error is modeled as a zero-mean, three-dimensional Gaussian error, with a 95% probability that the magnitude of the displacement is less than 1.0 mm.
- The clinical requirement is that, for a needle insertion depth of 100 mm, the needle tip must remain within 3.0 mm of the target with 95% confidence.

Design and implement the simulation. Model the target displacement as a zero-mean, three-dimensional Gaussian error. Starting from zero error (sigma = 0), gradually increase the standard error (sigma) and determine the maximum value that can be tolerated while still satisfying the clinical requirement. Explain your methodology and findings.

Task	Subcomponent	Grading Criterion	Points	check sum
Tool Calibration	Calibration formulation	Clearly formulate the calibration problem: determine the tool-tip position and tool-axis direction in the marker-defined tool model frame from optical-tracker observations.	3	
	Tool-tip calibration method	Implement a mathematically correct method for recovering the tool-tip position from multiple tracked poses.	5	
	Tool-tip motion constraints	Use and explain appropriate motion constraints for tip calibration poses.	3	
	Tool-axis calibration method	Implement a mathematically correct method for recovering the tool-axis direction from tracked observations.	5	
	Tool-axis motion constraints	Use and explain appropriate motion constraints for axis calibration and sufficiently excite the relevant degrees of freedom.	3	
	Ground-truth simulation	Generate suitable simulated tool motions and corresponding marker positions from a known ground-truth tool model.	3	
	Perfect-data validation	With exact marker observations, demonstrate recovery of the ground-truth tool-tip position and tool-axis direction to numerical precision.	2	
	Gaussian marker-error simulation	Add zero-mean 3-D Gaussian marker-localization error	3	
	Noisy calibration results	Repeat calibration with noisy observations; quantify tool-tip position error and tool-axis direction error and explain the findings.	2	
	Implementation / explanation	Provide clear code structure, comments, and/or supporting explanation; implement appropriate reusable utilities as separate modules.	1	30
Registration	Rigid registration	Correctly determine the rigid transformation from the three marker coordinates in the tool model frame to the corresponding marker coordinates in the CT frame.	7	
	Transform tool tip	Correctly compute the tool-tip position Pct in the CT frame.	3	
	Transform tool axis	Correctly compute the tool-axis direction Vct in the CT frame, treating a direction vector appropriately under rigid transformation.	3	
	Pure-translation test	Construct a known pure-translation test case, predict the expected result on paper, and verify the computed registration.	2	
	Pure-rotation test	Construct a known pure-rotation test case, predict the expected result on paper, and verify the computed registration.	2	
	Rotation + translation test	Construct a known test case containing both rotation and translation, predict the expected result, and verify the computed registration.	2	
	Explanation / numerical agreement	Clearly explain the method and demonstrate agreement between predicted and computed results.	1	20

Biopsy Plan	Targeting error	Correctly compute the shortest distance from the needle axis to the tumor center.	5	
	Tissue-core length	Correctly compute the length of the needle trajectory contained within the spherical tumor model.	5	
	Needle insertion depth	Correctly compute the required insertion depth for harvesting the predicted tissue core.	5	
	Intersecting trajectory test	Construct a simple ground-truth case in which the needle intersects the tumor and verify all expected outputs.	1	
	Tangent trajectory test	Construct a simple ground-truth case in which the needle just touches the tumor and verify the expected outputs.	1	
	Miss trajectory test	Construct a simple ground-truth case in which the needle entirely misses the tumor and verify the expected outputs.	1	
	Geometry utilities / explanation	Use appropriate reusable geometry utilities (e.g., point-to-line distance and line-sphere intersection) and clearly explain assumptions and edge cases.	2	20
Targeting Error Simulation	Ideal scene / baseline	Set up the ideal scene consistently with the assignment	2	
	Marker-localization error model	Model independent marker-localization error	4	
	Error propagation through registration	Re-register the tool from noisy marker observations so that marker-localization error correctly propagates into tool-tip and tool-axis estimates.	5	
	Target-displacement model	Model target displacement	4	
	Compound targeting error	Correctly compute the distance between the resulting needle trajectory and the displaced target point.	4	
	Statistical design	Explain a sufficiently large number of independent trials and a sound simulation procedure	3	
	Sigma sweep	Starting from sigma = 0, design and complete the sweep	3	
	Maximum tolerable sigma	Determine and justify the maximum sigma for which the clinical requirement is met	3	
	Results and discussion	Clearly present results with correct units/labels and explain the findings, interaction of error sources, statistical uncertainty, and limitations.	2	30
			100	**100**